CONFIDENTIAL TECHNICAL REPORT

Maintenance & Health Check (MHC) Report

System Health, Calibration & Optical Performance Assessment

MACHINE & CUSTOMER IDENTITY | **MHC EVALUATION & ASSESSMENT**

CUSTOMER ACCOUNT **Acme PCB Corp**	PLANT / FACILITY **Plant 2 - Penang**
PRODUCTION LINE —	ZONE —
MACHINE MODEL **ESI 5330 Flex**	MACHINE NUMBER / SOURCE **WLVIA#3**
SERIAL NUMBER **SN-98765**	
BASELINE DATE —	LAST MHC DATE 2026-09-02

SERVICE ENGINEER	INSPECTION DATE	RELEASE STATUS
Sahafiz Field Service Engineer	2026-09-02	PENDING REVIEW

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REPORT INDEX

14 Subsystem Diagnostics & Certification Modules

MAINTENANCE & HEALTH CHECK REPORT STRUCTURE

This formal technical report presents authoritative inspection, telemetry, and calibration records collected for this equipment. Subsystems are documented with baseline comparisons and recorded evidence where applicable.

03 EXECUTIVE SUMMARY

MHC EVALUATION

MHC RESULT

ACTION REQUIRED — 33%

ACTION_REQUIRED

Routine Maintenance & Health Check (MHC) completed for Flex Drill WLVIA#3 (SN-98765) at Acme PCB Corp - Plant 2 - Penang.

MHC INSPECTION RESULTS

SUBSYSTEM / AUDIT ITEM	SPECIFICATION	VERDICT
Laser Power (Laser Head 1 & 2)	15.0 W ± 10% (13.5–16.5 W)	WARNING
Beam Profile / Mode	–	PASS
Stage Calibration	–	— NOT COLLECTED
AGC / Scanner Calibration	–	— NOT COLLECTED
Temperature & Cooling	–	PASS

04 LASER LIFECYCLE & USAGE TELEMETRY

1 HEADS ANALYZED

Laser Head 1

PASS

OPERATING RUN TIME	REMAINING LIFESPAN	WARNING THRESHOLD	RATED EOL LIMIT
14,250 hrs	5,750 hrs	18,000 hrs	20,000 hrs
Accumulated Run Time	28.7% Capacity Left	Maintenance Alert	Total Tube Lifespan

0 hrs — 5,750 HOURS REMAINING (28.7%) — 20,000 hrs (EOL)

EST. REMAINING DAYS (24/7 PACE)	PROJECTED EOL DATE	TELEMETRY VERIFICATION
239 days (~0.7 yrs)	2027-04-29	Physical Counter Verified

LIFECYCLE PROGNOSIS & SERVICE RECOMMENDATIONS

PROGNOSTIC ADVISORY

Laser Head 1: Mid-to-late life phase (28.7% remaining). Tube capacity is stable. Monitor optical power decay during regular service.

05 LASER POWER & BASELINE COMPARISON

Authoritative Optical Power Measurement & Calibration Variance Analysis

SPEC: 15.0W ± 10% (13.50–16.50 W)

COMPLETE

LEFT: Historical Baseline (Previous) VS **RIGHT:** Present Measurement (Current) → **VARIATION:** Δ = Current − Previous

Laser Head 1

SPEC: 15.00 W ± 10% (13.50–16.50 W)

PASS

PREVIOUS (BASELINE)			CURRENT (PRESENT)
Baseline			2026-09-02
— Historical Baseline	VS COMPARE		15.10 W Target: 15.00 W

VARIATION = CURRENT − PREVIOUS Calibration Shift: **No previous baseline**

Laser Head 2

SPEC: 15.00 W ± 10% (13.50–16.50 W)

WARNING

PREVIOUS (BASELINE)			CURRENT (PRESENT)
Baseline			2026-09-02
— Historical Baseline	VS COMPARE		Not Recorded Target: 15.00 W

VARIATION = CURRENT − PREVIOUS Calibration Shift: **No previous baseline**

06 OPTICAL BEAM PROFILE & SPOT QUALITY

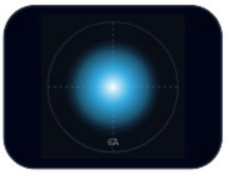
Spatial Beam Distribution & Multistage Aperture Quality Telemetry

PASS

LASER HEAD 1 Laser Head 1

Baseline: 3.500 mm | Δ Variation: +0.000 mm (0.0%) PASS

1. LASER SOURCE (6A) PASS

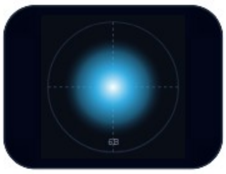


CURRENT MEASURED SPOT SIZE

Ø 3.520 mm

APPLICABLE SPECIFICATION:
3.5 mm ±10% (3.15–3.85 mm)

2. FLAT TOP (6B) PASS



CURRENT MEASURED SPOT SIZE

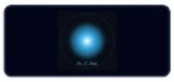
Ø 4.150 mm

APPLICABLE SPECIFICATION:
4.2 mm ±5% (3.99–4.41 mm)

APERTURE MASK PROFILE EVIDENCE (0.9 MM – 2.2 MM)

6 / 6 Within Target Spec

0.9mm PASS

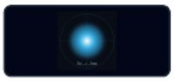


Ø 0.950 mm
≥0.9mm

1.1mm PASS

Ø 1.150 mm
≥1.1mm

1.3mm PASS

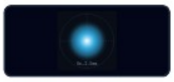


Ø 1.350 mm
≥1.3mm

1.8mm PASS

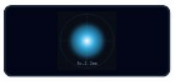
Ø 1.850 mm
≥1.8mm

2.0mm PASS



Ø 2.050 mm
≥2.0mm

2.2mm PASS

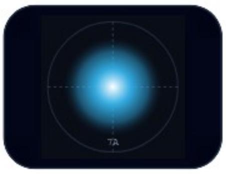


Ø 2.250 mm
≥2.2mm

LASER HEAD 2 Laser Head 2

Baseline: 3.500 mm | Δ Variation: +0.000 mm (0.0%) PASS

1. LASER SOURCE (7A) PASS

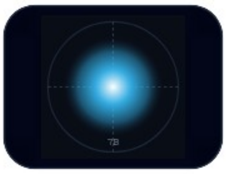


CURRENT MEASURED SPOT SIZE

Ø 3.480 mm

APPLICABLE SPECIFICATION:
3.5 mm ±10% (3.15–3.85 mm)

2. FLAT TOP (7B) PASS



CURRENT MEASURED SPOT SIZE

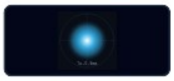
Ø 4.150 mm

APPLICABLE SPECIFICATION:
4.2 mm ±5% (3.99–4.41 mm)

APERTURE MASK PROFILE EVIDENCE (0.9 MM – 2.2 MM)

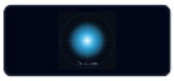
6 / 6 Within Target Spec

0.9mm PASS



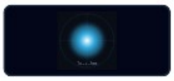
Ø 0.950 mm
≥0.9mm

1.1mm PASS



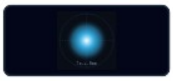
Ø 1.150 mm
≥1.1mm

1.3mm PASS



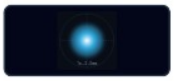
Ø 1.350 mm
≥1.3mm

1.8mm PASS



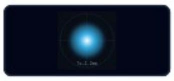
Ø 1.850 mm
≥1.8mm

2.0mm PASS



Ø 2.050 mm
≥2.0mm

2.2mm PASS



Ø 2.250 mm
≥2.2mm

07 FOCUS OPTIMIZATION

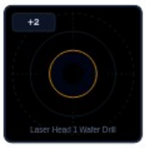
Machining Focus Calibration & Focal Sequence Verification

FOCUS ADJUSTMENT DATE	ADJUSTMENT REASON	BASELINE	EVALUATION
2026-09-02	Laser source replacement	-0.300 mm	Laser source optical alignment verified via dummy wafer drill matrix. Focus set to 0.00 position.

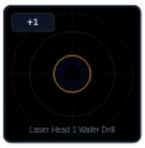
LASER HEAD 1

BASELINE: -0.300 mm

+0.300 mm
Focus Position



+0.200 mm
Focus Position



+0.100 mm
Focus Position

0.000 mm
Focus Position

-0.100 mm
Focus Position

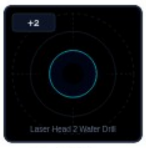
-0.200 mm
Focus Position

-0.300 mm
Focus Position

LASER HEAD 2

BASELINE: -0.300 mm

+0.300 mm
Focus Position



+0.200 mm
Focus Position

+0.100 mm
Focus Position

0.000 mm
Focus Position

-0.100 mm
Focus Position

-0.200 mm
Focus Position

-0.300 mm
Focus Position

Note: Top via impact: Focus adjustment primarily affects top diameter (~90%), significantly influencing top via.

08 POWER OFFSET & CALIBRATION

Power Offset & Process Optimization

Power Offset Range: −20% to +20%

LASER HEAD 1 (LH1)

POWER OFFSET: —

	PHASE 1	PHASE 2
Recipe:	–	–
Adjusted:	**–**	**–**

OFFSET COMPARISON — Prev: – → Curr: –

ADJUSTMENT REASON

—

LASER HEAD 2 (LH2)

POWER OFFSET: —

	PHASE 1	PHASE 2
Recipe:	–	–
Adjusted:	**–**	**–**

OFFSET COMPARISON — Prev: – → Curr: –

ADJUSTMENT REASON

—

Note: Bottom via impact: Power offset primarily influences bottom via diameter (~90%), making it a key factor in bottom via diameter control.

09 STAGE CALIBRATION (X/Y DEVIATION)

Sub-Micron Motion Stage Positional Accuracy & Cross-Axis Verification

TOLERANCE: ±2.0 µm

— NOT COLLECTED

STAGE IDENTIFIER	X DEVIATION RANGE [MEASURED]	Y DEVIATION RANGE [MEASURED]	MAX ABS DEV [DERIVED]	VERDICT
Stage 1 (Left)	–	–	**–**	— UNANSWERED
Stage 2 (Right)	–	–	**–**	— UNANSWERED

Calibration Record: Stage positional calibration completed across the full travel range, with X/Y deviation verified against the defined tolerance.

10 AGC / SCANNER CALIBRATION

Galvo Scanner & Automatic Grid Calibration Positional Verification

TOLERANCE: ±3.0 µm

— NOT COLLECTED

AGC IDENTIFIER	X DEVIATION RANGE [MEASURED]	Y DEVIATION RANGE [MEASURED]	VERDICT
AGC 1	–	–	— UNANSWERED
AGC 2	–	–	— UNANSWERED

Calibration Record: AGC/scanner positional calibration completed across the full travel range, with X/Y deviation verified against the defined tolerance.

11 TEMPERATURE & THERMAL TELEMETRY

COMPLETE

Optics Markbox 6-Channel Air-Cooling & Thermal Telemetry Monitoring

OPTICS MARKBOX AIR-COOLING SUBSYSTEM OVERVIEW TARGET SPEC: —

MARKBOX 1	MARKBOX 2	MARKBOX 3
NO DATA	NO DATA	NO DATA
Not Linked	**Not Linked**	**Not Linked**
Optics Markbox 1 (CH1 + CH4)	Optics Markbox 2 (CH2 + CH5)	Optics Markbox 3 (CH3 + CH6)

NO CONTINUOUS MULTI-CHANNEL THERMAL LOG LINKED TO THIS SESSION

To populate detailed 6-channel optics markbox sensor matrices and profile curves, link a temperature log record to this machine.

ENGINEER OBSERVATION: Cooling optimal

12 PRODUCT PROCESS & VIA QUALITY

Product Recipe Parameters & Via Measurements

COMPLETE

PRODUCT	RECIPE / PROGRAM	LOT / PANEL IDENTIFIER
Flex Rigid Standard	REC-01	SMP-100

PROCESS RECIPE PHASE PARAMETERS

PROCESS PHASE	POWER (W)	FREQUENCY (kHz)	SHOT COUNT	MASK (mm)	DEFOCUS (mm)
Phase 1	-	-	-	-	-
Phase 2	-	-	-	-	-

VIA DRILLING MEASUREMENTS & SPECIFICATION

LASER HEAD	TOP DIA.	BOT DIA.	TAPER	VERDICT
Acceptance Spec	—	—	—	Engineering Spec
Laser Head 1 (LH1)	50.0 µm	-	-	PASS
Laser Head 2 (LH2)	-	-	-	PASS

VIA CROSS-SECTION EVIDENCE

No Image

Laser Head 1 (LH1)

No Image

Laser Head 2 (LH2)

Engineer Remarks: Profile OK

13 FINDINGS & OBSERVATIONS

TOTAL RECORDED: 0

— NOT COLLECTED

—

ALL SUBSYSTEMS NOMINAL

14 SPARE PARTS & RECOMMENDATIONS

COMPLETE

ENGINEERING RECOMMENDATIONS & FUTURE ACTION PLAN

—

CONSUMED PARTS (SERVICE EXECUTION)

PART NAME	QTY	ACTION	COSTING
Air Filter Element	1	REPLACED	CUSTOMER_COST

RECOMMENDED SPARE PARTS (PROCUREMENT / STOCK)

—

15 BUYOFF & OFFICIAL APPROVALS

PRODUCTION RELEASE: PENDING

PENDING REVIEW

NEXT SCHEDULED MHC Target Due Date: **2026-12-01** Cycle: **90-Day Standard Interval**

FIELD SERVICE ENGINEER VERIFIED

Sahafiz
Field Service Engineer
Date: 2026-09-02

[COMPLETED & SIGNED BY ENGINEER]

CUSTOMER ACCEPTANCE REPRESENTATIVE PENDING

Pending Customer Sign-off
Acme PCB Corp
Date: —

[PENDING CUSTOMER REVIEW & SIGN-OFF]